

JMP Exam Notes

ONE MEAN • POWER • TWO PROPORTIONS

➤ = then click • ▼ = the little red triangle next to that report's title • → = where the answer shows up • reject H_0 when $p \leq \alpha$

GETTING STARTED

Open the data. File ➤ Open ➤ the .jmp file. Number columns have to say **Continuous** — if one doesn't, right-click the column heading ➤ Column Info ➤ Modeling Type. Typing data in yourself: File ➤ New ➤ Data Table, double-click a column heading to rename it, type the values down the rows, double-click to the right of the last column to add another.

Is it a number or a group? If you could take an average of it, it's a number → use a **mean**. If it just sorts each person into a group (yes/no, accepted/rejected), it's a group → use a **proportion**.

Writing the hypotheses. H_0 always says μ equals the claimed number. "less / went down" → $H_1: \mu < \cdot$ "more / went up" → $> \cdot$ "different / changed" → $\neq \cdot$. Use μ (the true mean for everyone), never $\bar{x}$ (your sample's mean).

Shape, average, spread, count. Analyze ➤ Distribution ➤ Y, Columns = variable ➤ OK → the histogram shows the shape; **Summary Statistics** gives Mean, Std Dev, N and Std Err Mean (how much a sample average typically bounces around).

TESTING ONE MEAN

t test — when they don't give you σ . Analyze ➤ Distribution ➤ Y = variable ➤ OK ➤ ▼variable ➤ Test Mean ➤ type the claimed number ➤ leave the standard deviation box empty ➤ OK → the **Test Mean** panel, **t Test** row.

z test — when they do give you σ . Same path, but also fill in "Enter True Standard Deviation to do z-test rather than t test" → the panel now shows a **z Test** row instead.

Which p-value to read. H_1 has $\neq \rightarrow \text{Prob}>|t|$ • H_1 has $> \rightarrow \text{Prob}>t$ • H_1 has $< \rightarrow \text{Prob}<t$. Ignore the Signed-Rank line.

Only summary numbers given. Not in JMP — it needs the raw data. Use $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$, $df = n - 1$. If they give you s^2 , take the square root first to get s .

"1.03 standard errors below." That sentence already is the answer: $t = -1.03$. Below means negative, above means positive.

WHAT HAS TO BE TRUE, AND INTERVALS

Conditions for the t test. Independence: it's a random sample, it's less than 10% of the whole group, and each person is measured once. **Bell shape:** the problem says bell-shaped, **or** $n \geq 30$ (big samples are fine either way), **or** the histogram looks even with no far-out points.

Checking the shape in JMP. ▼variable ➤ Continuous Fit ➤ Normal ➤ ▼Fitted Normal ➤ Goodness of Fit → **Shapiro-Wilk Prob<W**; if it's bigger than α , the bell shape is fine. Also ▼variable ➤ Normal Quantile Plot — the dots should hug the straight line.

Confidence interval for the mean. **Summary Statistics** already lists Upper / Lower 95% Mean. For another level: ▼variable ➤ Confidence Interval ➤ 0.90 / 0.95 / 0.99. Formula $\bar{x} \pm t(\alpha/2, n-1) \cdot s/\sqrt{n}$.

Chance an average lands past a number. $z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$, or Cols ➤ New Column ➤ Formula ➤ Probability ➤ Normal Distribution($a, \mu, \sigma/\sqrt{n}$), which gives the chance of landing **below a** — subtract from 1 for above.

IF H_0 WERE TRUE, AND POWER

Describe the spread of sample averages when H_0 is true. Give all three. **Shape:** bell-shaped, because the population is bell-shaped or because $n \geq 30$. **Center:** μ_0 , the number H_0 claims. **Spread:** $\sigma/\sqrt{n}$.

Which sample averages make you reject? cutoff = $\mu_0 + (z)(SE)$, where $SE = \sigma/\sqrt{n}$ and z carries the sign of the tail: -1.645 for a "less than" test, $+1.645$ for a "greater than" test at $\alpha = .05$; for a two-sided test use $\mu_0 \pm 1.96 \cdot SE$. Reject when $\bar{x}$ is past that cutoff.

β — the chance of missing a real difference. Take that same cutoff, but measure it from the **true** mean instead: $z = (\text{cutoff} - \mu_{\text{true}}) / (\sigma/\sqrt{n})$. Then β is the chance $\bar{x}$ lands in the "don't reject" zone.

Power = $1 - \beta$. These all mean power: "chance of detecting this difference", "chance of rejecting H_0 when the true mean is $\cdot$ ", "chance of correctly concluding". These mean β : "chance of failing to reject", "chance of a Type II error".

Worked example. $\mu_0 = 50$, $\sigma = 21$, $n = 36$, $\alpha = .05$, "less than" test. $SE = 21/6 = 3.5$; cutoff = $50 - 1.645(3.5) = 44.24$. If the true mean is 43: $z = (44.24 - 43)/3.5 = .354$, $\beta = .3613$, power = **.6387**.

Power from JMP. DOE ➤ Design Diagnostics ➤ Sample Size and Power ➤ One Sample Mean, fill in Alpha, Std Dev, Difference to Detect and Sample Size, leave **Power** empty ➤ Continue. Leaving **Sample Size** empty instead tells you how big a sample you'd need. It assumes a two-sided test, so on one-sided questions use the formula above.

Drawing it. Two bell curves on one line. Top one centered at μ_0 , with the tail past the cutoff shaded — that's α . Bottom one centered at the true mean, with the "don't reject" side shaded — that's β , and the rest is power. One vertical line at the cutoff through both.

TWO PROPORTIONS — SEPARATE GROUPS

Typing in a table of counts. File ➤ New ➤ Data Table with three columns — Group (Nominal), Outcome (Nominal), Count (Continuous) — and one row per box of the table, like (Help, Accepted, 222).

Analyzing it. Analyze ➤ Fit Y by X ➤ Y = Outcome ➤ X = Group ➤ Freq = Count ➤ OK → **Contingency Table** and **Tests**. Skip the **Freq** box and JMP thinks you only have 4 people.

Best guess for the difference. ▼Contingency Table ➤ Row %. Each $\hat{p}$ is the count divided by **that group's own row total**, not the overall total. Example: $222/515 = 55/190 = .4311 - .2895 = .1416$.

Hypotheses. $H_0: p_1 = p_2 = 0$ — if both groups have the same rate, the difference is zero. "improves" → > 0 • "reduces" → < 0 • "different" → $\neq 0$.

What has to be true. Two separate random groups, or people randomly assigned to them; one answer per person; and all four counts in the table at least 5.

Test and interval. ▼Contingency Analysis ➤ Two Sample Test for Proportions → the difference, its standard error, **Prob>|z|** / **Prob>z** / **Prob<z**, and the confidence interval. **Pearson ChiSquare** in the Tests panel is the same two-sided test.

Formulas. combined $\hat{p} = (x_1+x_2)/(n_1+n_2)$; $z = (\hat{p}_1-\hat{p}_2) / \sqrt{[\hat{p}(1-\hat{p})(1/n_1+1/n_2)]}$. The interval doesn't combine them: $(\hat{p}_1-\hat{p}_2) \pm z\alpha/2\sqrt{[\hat{p}_1\hat{q}_1/n_1 + \hat{p}_2\hat{q}_2/n_2]}$.

TWO PROPORTIONS — SAME PEOPLE TWICE

Spotting it and reading the table. Same people measured twice, or matched pairs.

a and **d** are the people who didn't change; **b** and **c** are the ones who switched, one each way. Smoking: $a = 20$, $b = 5$, $c = 16$, $d = 9$, $n = 50$. Don't use the test above — it ignores the pairing.

Only the switchers count. $n' = b + c$. Everyone who stayed the same ($a + d$) gets dropped. $T =$ how many switched in the direction H_1 expects. Smoking: 29 dropped, $n' = 21$, $T = 16$. H_0 says switching is equally likely both ways.

The p-value. If H_0 is true, T acts like flipping a coin n' times: $T \sim \text{Binomial}(n', 0.5)$. Cols ➤ New Column ➤ Formula ➤ Discrete Probability ➤ Binomial Distribution(.5, n' , k) gives the chance of k or fewer. Smoking: $1 - \text{BinDist}(5, 21, 15) = .0133$, under .05, so the video worked.

Or let JMP do it. Analyze ➤ Fit Y by X ➤ Y = After, X = Before, Freq = Count ➤ OK ➤ ▼ ➤ Agreement Statistic → the **test of symmetry** is McNemar's test, but it only gives the two-sided answer.

Interval. $(b-c)/n \pm z\alpha/2\sqrt{[(a+d)(b+c) + 4bc] / n^3}$. Smoking: $(- .389, -.051)$, all below zero. Puzzle: $(-.196, .077)$, which includes zero — no real difference.

NUMBERS YOU MAY NEED TO LOOK UP

df: $n - 1$ for one mean • $n - 2$ for a slope • $n - k - 1$ for one predictor out of k • k and $n - k - 1$ for the overall F.

z cutoffs: one-sided $\alpha = .10 \rightarrow 1.282$, $.05 \rightarrow 1.645$, $.025 \rightarrow 1.960$, $.01 \rightarrow 2.326$, $.005 \rightarrow 2.576$. Two-sided $90\% \rightarrow 1.645$, $95\% \rightarrow 1.960$, $98\% \rightarrow 2.326$, $99\% \rightarrow 2.576$.

JMP formula functions: t Quantile($1-\alpha/2$, df) for a t cutoff, Normal Quantile($1-\alpha$) for a z cutoff, t Distribution(t, df) for the chance below t, $2*(1 - t \text{ Distribution}(|t|, df))$ for a two-sided p-value.

FILL-IN-THE-BLANK ANSWERS

DEFINE THE PARAMETER — μ = the true mean _____ (with units) for all _____.

INTERPRET THE TEST STATISTIC — The sample average is _____ standard errors [above / below] _____, the number H_0 claims.

INTERPRET THE P-VALUE — If the true mean really were _____, there is a _____ % chance of getting a sample average at least this far from _____ just by chance.

CONCLUSION — WORKS FOR EVERY TEST — Since $p =$ _____ [$< / >$] $\alpha =$ _____, we [reject / fail to reject] H_0 . There [is / is not] enough evidence to say that _____.

Last blank → **ONE MEAN:** — the mean [variable] is [less than / greater than / different from] _____. **TWO PROPORTIONS:** — the proportion of _____ is [higher / lower / different] for ____ than for _____. **SLOPE:** — [x] helps predict [y] • **OVERALL F:** — the model helps predict [y] • **CURVED MODEL:** — the squared term makes the model better. Never write "accept H_0 ." If your instructor words it "sufficient evidence to conclude," that is the same sentence.

INTERPRET ANY CONFIDENCE INTERVAL — We are _____ % confident that _____ is between _____ and _____.

Middle blank → **M:** — the true mean [variable] for all [group] • **$P_1 - P_2$:** — the difference in the proportion of _____ between _____ and _____ • **SLOPE:** — the average change in [y] for each extra one [x-unit] • **MEAN AT X:** — the average [y] for all [units] with $x =$ _____ • **PREDICTION:** — a single [unit] with $x =$ _____ will have [y]. If the interval includes 0 → no real difference / doesn't help predict.

TYPE I ERROR IN CONTEXT — We decide that _____ when really _____ . The chance of this is $\alpha =$ _____.

TYPE II ERROR IN CONTEXT — We decide there isn't enough evidence that _____ when really _____ is true. The chance of this is $\beta =$ _____.

DESCRIBE THE SPREAD OF SAMPLE AVERAGES — **Shape:** bell-shaped, because _____ . **Center:** $\mu =$ _____ . **Spread:** $\sigma/\sqrt{n} =$ _____ / $\sqrt{\text{_____}} =$ _____.

Reason blank: "the problem says the population is bell-shaped" or " $n \geq 30$, so it's fine either way".

CHECK THE T-TEST CONDITIONS — **Independence:** the _____ are a random sample and fewer than 10% of all _____ . **Bell shape:** _____.

POWER AND B — If the true mean is really _____, there is a _____ % chance we [correctly reject / fail to reject] H_0 . "Correctly reject" = power = $1 - \beta$. "Fail to reject" = β = the chance of a Type II error.

WHICH ERROR WAS POSSIBLE — We [rejected / failed to reject] H_0 , so if we made a mistake it has to be a Type [I / II] error: _____.

LARGE ENOUGH SAMPLE, TWO PROPORTIONS — The counts _____, _____, _____ and _____ are all at least 5, so both groups are big enough.

JMP Exam Notes

CORRELATION AND THE BEST-FIT LINE

Which variable goes where. "Predict A using B" → A is the response and goes in **Y, Response**; B does the predicting and goes in **X, Factor**.

Scatterplot. Analyze > Fit Y by X > Y, Response = response > X, Factor = factor > OK.

Add the line. ▾Bivariate Fit > Fit Line → equation printed right above **Summary of Fit**. For the curved bands around it: ▾Linear Fit > Confid Curves Fit or Confid Curves Indiv (picture only).

Correlation r. Analyze > Multivariate Methods > Multivariate > both variables in Y, Columns > OK → **Correlations**, the number where the two variables cross (the 1.0000s are each variable with itself). Or $r = \pm\sqrt{\text{RSquare}}$, using the sign of the slope. r always lands between -1 and $+1$: the sign is the direction, the size is the strength — $|r| \geq .8$ strong, $.5$ – $.8$ medium, under $.5$ weak.

Equation of the best-fit line. Analyze > Fit Model > Y = response > select x > Add > Run → **Parameter Estimates**, **Estimate** column: the Intercept row is b_0 , the x row is b_1 .

Predict. Put the x value into $\hat{y} = b_0 + b_1x$. Watch the units — if x is in thousands, 34,000 means $x = 34$. Don't predict past the range of the data you have.

CHECKING IT'S OK TO USE THE LINE

Save the leftovers. ▾Response y > Save Columns > Residuals, then again for **Predicted Values** and **Studentized Residuals** → new columns at the right of the data table. A residual is just actual — predicted.

Are the leftovers bell-shaped? Analyze > Distribution > Y = Residual > OK > ▾Residual > Normal Quantile Plot, then Continuous Fit > Normal > ▾Fitted Normal > Goodness of Fit → **Shapiro-Wilk**.

Leftovers plot. ▾Response y > Row Diagnostics > Plot Residual by Predicted. A shapeless band around zero is what you want · an arch or U means a line is the wrong shape · a funnel means the spread isn't even · a studentized value past ± 3 is a far-out point.

DOES X ACTUALLY HELP PREDICT Y?

Test the slope. Fit Model > Run > Parameter Estimates, on the x row → **t Ratio** is the test number, **Prob>|t|** is the two-sided p-value, **Std Error** is the slope's wobble. H_0 : the slope is 0, $df = n - 2$.

One-sided? Cut **Prob>|t|** in half, but only if the slope's sign matches what H_1 says. If it doesn't, the p-value is over .5 and you fail to reject.

Testing the slope against a number other than 0. JMP only tests 0. Either check whether the confidence interval contains that number, or use $t = (b_1 - c) / \text{Std Error}$, $df = n - 2$.

How much of y is explained. **Summary of Fit** → **RSquare**, times 100 for a percent.

THE THREE INTERVALS

Interval for the slope. Right-click inside **Parameter Estimates** > Columns > Lower 95% + Upper 95% → read them on the **x** row, not the Intercept row. If it doesn't include 0, x helps predict y.

99% or 98% instead. ▾Response y > Set Alpha Level > 0.01 (or **Other** → 0.02), then read those columns again. If yours is stuck at 95%: $b_1 \pm t(\alpha/2, n-2) \cdot \text{Std Error}$, with t from t Quantile($1-\alpha/2$, $n-2$).

Interval for the average y — "for ALL cases with $x = \underline{\hspace{1cm}}$ ". Add a row with that x and **leave Y empty** (that keeps it out of the line), then Fit Model > Run > ▾Response y > Save Columns > Mean Confidence Interval → **Lower / Upper 95% Mean** on that row.

Interval for ONE new case. Same setup, then Save Columns > Indiv Confidence Interval → **Lower / Upper 95% Indiv**, always wider. A singular word — "a truck", "a school" — means use Indiv.

Far-out points. Click the point to light up its row, then Rows > Exclude/Unexclude and Run again. "How many more votes than predicted" = actual — predicted, which is that row's residual.

SEVERAL PREDICTORS AT ONCE

Look at them first. Analyze > Multivariate Methods > Multivariate > response + all predictors in Y, Columns > OK → **Correlations** and a grid of mini scatterplots. Two predictors with $|r|$ above about .8 measure nearly the same thing, which causes trouble.

Fit it. Analyze > Fit Model > Y = response > highlight all the predictors > Add > Run → **Parameter Estimates** gives $\hat{y} = b_0 + b_1x_1 + \dots + b_kx_k$.

Does the whole model help? Analysis of Variance → F Ratio and Prob > F. H_0 says every slope is 0. Rejecting means at least one predictor helps — not that they all do.

Does one predictor help, with the others in? **Parameter Estimates**, that predictor's row → **t Ratio**, **Prob>|t|**, $df = n - k - 1$. A predictor can look strong alone and still be useless here, because another one already covers it.

Interval for one coefficient. Right-click **Parameter Estimates** > Columns > Lower 95% + Upper 95%; change the level first with **Set Alpha Level**.

Checking conditions. Save **Residuals**, use ▾Response y > Row Diagnostics > Plot Residual by Predicted, and run Distribution on the residuals for the plot and Shapiro-Wilk. For overlap between predictors: right-click **Parameter Estimates** > Columns > VIF — over 5 a warning, over 10 a problem.

PICKING THE BEST MODEL

Use adjusted R². **Summary of Fit** → **RSquare Adj**, highest one wins. Plain R^2 always goes up when you add a predictor, so it can't compare models with different numbers of predictors; $R^2_{\text{adj}} = 1 - (1 - R^2)(n-1)/(n-k-1)$ takes a penalty for extras.

Every model at once. Fit Model > add all > Personality = Stepwise > Run > ▾ > All Possible Models → every combination with RSquare, RSquare Adj, Cp, AICc, BIC.

Dropping predictors one at a time. Fit Model > add all > Stepwise > Run, set Direction = **Backward**, Stopping Rule = **P-value Threshold**, Prob to Leave = **0.1**, click **Go** → **Current Estimates**, a check under **Entered** means it stayed. Then **Make Model** → **Run**. The idea: throw out the biggest p-value above 0.1, refit, repeat.

CORRELATION · ONE PREDICTOR · SEVERAL PREDICTORS · CURVED

Predicting from the model you picked. Put the values into that model's equation, or add a row with those x's and an empty Y and save **Predicted Values**, or use ▾Response y > Factor Profiling > Profiler.

CURVED (QUADRATIC) MODELS

Quick way. Analyze > Fit Y by X > Y, X > OK > ▾Bivariate Fit > Fit Polynomial > 2, quadratic → a **Polynomial Fit Degree=2** block with the plain equation printed above it.

Full way (use this one for intervals). Analyze > Fit Model > Y > select x > Macros > Polynomial to Degree with Degree = 2 > Run → **Parameter Estimates** gains an **x*x** row.

JMP shifts x first. By default it prints $(x-\bar{x}) \cdot (x-\bar{x})$ instead of x^2 , so the numbers aren't the plain b_0, b_1, b_2 . Uncheck ▾Response y > Center Polynomials and Run again to get the plain equation.

Does the curve help? Test the squared term only: **Parameter Estimates**, the **x*x** row (or **Effect Tests**) → **Prob>|t|**. H_0 says the squared term is 0, meaning a straight line is enough. Don't use the overall F here — it can look good just from the straight-line part.

The change in y isn't the same every time. change = $b_1(x_2 - x_1) + b_2(x_2^2 - x_1^2)$. Going 0 to 1 gives $b_1 + b_2$; going 2 to 3 gives $b_1 + 5b_2$. They're different on purpose — that's the whole point of asking both.

Intervals from a curved model. Add the row with x and an empty Y, fit with **Polynomial to Degree 2**, then **Save Columns** → **Mean Confidence Interval** for the average of all such cases, **Indiv Confidence Interval** for one specific new case.

THE FORMULAS BEHIND THE LINE

Three building blocks. $\bar{x}$ and $\bar{y}$ are the two averages, **n** is how many (x, y) pairs, **Σ** means add it up over every pair. JMP never shows these three — they're only for working a small data set out yourself. $S_{xx} = \Sigma(x-\bar{x})^2$ spread of the x's, feeds the slope and its standard error · $S_{yy} = \Sigma(y-\bar{y})^2$ spread of the y's, feeds r and R^2 · $S_{xy} = \Sigma(x-\bar{x})(y-\bar{y})$ how much x and y move together, feeds the slope and r.

Everything else comes from those three. slope $b_1 = S_{xy}/S_{xx}$ · intercept $b_0 = \bar{y} - b_1\bar{x}$ (needs b_1 first) · the line $\hat{y} = b_0 + b_1x$ · $r = S_{xy}/\sqrt{S_{xx} \cdot S_{yy}}$ · $R^2 = r^2$ · one point's leftover $e = y - \hat{y}$. To check your work, your b_0 and b_1 should match **Parameter Estimates**.

FILL-IN-THE-BLANK ANSWERS — REGRESSION

COMMENT ON THE SCATTERPLOT —There is a [**positive / negative**], [**weak / medium / strong**], [**straight / curved**] relationship between _____ and _____.
Add if needed: "There is an odd point at ____." If it curves, say a squared term should be tried.

INTERPRET R — $r = \underline{\hspace{1cm}}$ means a [**weak / medium / strong**] [**positive / negative**] straight-line relationship between _____ and _____.
WRITE THE EQUATION —Predicted _____ = _____ + _____ (_____).

INTERPRET THE SLOPE —For every extra one _____, the predicted (average) _____ [**goes up / goes down**] by _____.
INTERPRET THE INTERCEPT —When _____ is 0, the predicted _____ is _____.
If $x = 0$ is outside the data: add "but $x = 0$ isn't in the range of the data, so this doesn't really mean anything."

INTERPRET R² — _____ % of the changes in _____ are explained by the [**straight / curved**] relationship with _____.
DOES X HELP PREDICT Y? —Since $p = \underline{\hspace{1cm}}$ [**< / >**] $\alpha = \underline{\hspace{1cm}}$, we [**reject / fail to reject**] H_0 : the slope is 0. There [**is / is not**] enough evidence to say that _____ helps predict _____.
INTERVAL FOR THE SLOPE —We are _____ % confident that for every extra one _____, the average _____ changes by between _____ and _____.
INTERVAL FOR THE AVERAGE Y —We are _____ % confident that the average _____ for all _____ with _____ = _____ is between _____ and _____.
INTERVAL FOR ONE NEW CASE —We are _____ % confident that a single _____ with _____ = _____ will have a _____ between _____ and _____.
WHAT THE LEFTOVERS PLOT SHOWS —The plot shows _____, so the [**straight-line / even spread**] condition [**is / is not**] met. Shapiro-Wilk gives $p = \underline{\hspace{1cm}}$ [**> / <**] α , so the bell shape [**is / is not**] reasonable.
First blank: "a shapeless band around zero" · "an arched pattern" · "a funnel shape".

INTERPRET A COEFFICIENT (SEVERAL PREDICTORS) —For every extra one _____, the predicted _____ [**goes up / goes down**] by _____, while the other predictors stay the same.

DOES THE WHOLE MODEL HELP? —Since $p = \underline{\hspace{1cm}}$ [**< / >**] $\alpha = \underline{\hspace{1cm}}$, we [**reject / fail to reject**] H_0 : every slope is 0. There [**is / is not**] enough evidence to say that the model helps predict _____.
ONE PREDICTOR, OTHERS ALREADY IN —With the other predictors already in the model, there [**is / is not**] enough evidence to say that _____ helps predict _____ (t = _____, $p = \underline{\hspace{1cm}}$).
PICKING A MODEL —The model with _____ has the highest adjusted R^2 (_____), so it's the one to use.
DOES THE CURVE HELP? —The squared term has $p = \underline{\hspace{1cm}}$ [**< / >**] $\alpha = \underline{\hspace{1cm}}$, so there [**is / is not**] enough evidence to say that the curved model fits better than a straight line.